

Portfolio Blueprint

A premium, heavily animated, config-driven AI engineer portfolio: five screens that move a visitor from attention to connection, with a grounded portfolio assistant that costs nothing to run. No implementation yet — this is the contract we build against.

STACK

Next.js 15 · TS · Tailwind v4

ASSISTANT

In-browser answer engine

RUNNING COST

₹0 / \$0 – forever

ESTIMATE

~8 working days

BLOCKERS

4 open decisions

WHAT CHANGED SINCE REV. A

- **The reference image is now styling-only.** You’ve confirmed its data was generated placeholder text. The résumé is the sole source of truth — this closes three of the six open decisions outright.
- **The assistant no longer calls any paid API.** Rewritten in §05 as an in-browser retrieval and answer-composition engine. Zero cost, zero keys, zero rate limits, and it cannot hallucinate. Removes the server route, secrets layer, spend caps, and abuse hardening entirely.
- **Motion is promoted from polish to architecture.** §06 is now a full motion system spec, including a real rotating neural-brain hero rather than a spinning flat image.

00 VERDICT

The six calls that shape everything else

-
- 01 **Content is data, not markup.** Every fact about you lives in one typed, schema-validated layer. Components receive props; they never contain a name, date, email, or URL. This is the literal reading of your “never hard code anything” rule, and it is what makes the rest possible.
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- 02 **The résumé is the only source of truth.** The reference image contributes look — palette weight, scale, spacing, the brain motif, the wave — and nothing else. Every number, name, date, and address comes from the CV.
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- 03 **The assistant runs entirely in the browser and costs nothing.** A real retrieval engine (BM25 index + intent routing + answer composition) over your portfolio data. No API, no key, no server, no subscription, no free tier to expire. It answers in under 50 ms, works offline, and is *structurally incapable* of inventing a fact.
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- 04 **Motion is a first-class feature, not a final coat of paint.** A generated 3D neural brain rotating slowly in the hero, scroll-linked section choreography, magnetic interactions, and a living contact wave — all compositor-only, all with genuine reduced-motion alternatives.
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- 05 **Total running cost is zero.** Static site, free hosting tier, no database, no API, no CMS, no analytics vendor. The only optional expense in the entire project is a custom domain, and even that has a free alternative. Full ledger in §07.
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- 06 **Ship in seven phases, each independently deployable.** A complete, animated, shippable portfolio exists at the end of Phase 4. The assistant is Phase 5 — additive, not load-bearing.
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01 SOURCE
OF
TRUTH

Which source governs what

Now that the mockup’s data is confirmed placeholder, the rule is clean: **the image governs appearance, the CV governs facts.** Nothing from the image enters the content layer.

Dark luminous ground · large display type · generous whitespace · five-section rhythm · the neural-brain hero motif · the contact-section wave · nav and footer structure

Name, role, location, email · both internships with real dates · all four projects with real stacks · every skill and tool · both competition placements · education and CGPA · interests

THE FIGURES WE WILL ACTUALLY SHIP

MOCKUP CLAIMED	REALITY PER CV	WHAT GOES ON THE PAGE
2+ years experience	Feb 2026 → present	6 months, two internships
15+ projects	4 documented	4 shipped projects
3 hackathons won	Top-12 IndiaSkills (Cloud); 5th, VISAI 2026	2 competition placements
10+ technologies	Countable from the CV skills list	The real count, computed from content
“Meeting Bot”	No CV entry — placeholder	Dropped
mdfaiz44@gmail.com	faiz31807@gmail.com	The CV address
Timeline 2023 / 2024	India Space Lab Feb–Mar 2026; Metawurks AI Apr 2026–present	Real dates

ONE DERIVED-NOT-TYPED DETAIL

The stat counters compute themselves. “4 shipped projects” is `projects.length` ; the technology count is the deduplicated union of every project’s stack; the experience duration is derived from the earliest start date. You add a fifth project and the hero updates itself — and it can never drift out of sync with the section below it.

THE ONE PLACE THE IMAGE CONTRADICTS YOUR BRIEF

Your brief says “*avoid excessive cards, crowded sections, or dashboard-like layouts.*” The image’s section 02 is five cards with percentage bars — a dashboard. I plan to follow your written brief: skills become grouped capability clusters, each linking to the project that proves it. “Artificial Intelligence — 90%” is a number you assigned yourself; “RAG pipelines, agentic systems, MCP servers → *see Newsroom AI*” is verifiable. The schema keeps an optional `level` field, so the bars are one flag away if you disagree. This is decision 1 in §08.

What “never hard code anything” means in practice

Taken literally the rule is unbuildable — some string has to exist somewhere. Taken usefully: **every value has exactly one home, and that home is chosen by what makes the value change.** With the paid API gone, the secrets layer disappears entirely — six layers, not seven.

LAYER	HOLDS	CHANGES WHEN	ENFORCED BY
L1 Content	Your facts — profile, experience, projects, skills, achievements, education, links	Your career changes	Zod schemas; import throws on invalid data
L2 Tokens	Colour, type scale, spacing, radii, easing curves, animation durations	The design changes	Tailwind v4 <code>@theme</code> ; lint bans arbitrary hex in class names
L3 Motion	Every duration, delay, stagger, spring constant, particle count, rotation speed	The feel is tuned	A single <code>motion.ts</code> config; no magic numbers inside components
L4 Environment	Site URL, feature flags, build mode	The deploy target changes	One <code>env.ts</code> that Zod-parses <code>process.env</code> ; nothing else may read it
L5 Structure	Section order, ids, nav labels, journey stage	You reorder the story	A single ordered <code>sections</code> registry that generates nav, scroll-spy and footer
L6 Derived	SEO metadata, OG image, sitemap, JSON-LD, stat counters, the assistant’s search index, its suggested questions	Never directly — computed from L1	Pure functions and a build step; no hand-written duplicates

THE DISTINCTION THAT MATTERS

Your name and email belong in **L1 content**, not L4 environment. They are versioned facts about you, not deployment variables — they should live in git with a review trail. Environment is only for what genuinely differs between local, preview, and production. Putting identity in env vars is a common over-correction that makes the project harder to run, not more configurable.

L3 is new in this revision and earns its place: an animated site accumulates magic numbers faster than anything else. `0.6`, `0.09`, `[0.16, 1, 0.3, 1]` scattered across twenty components is exactly the hardcoding you asked me to avoid, and it makes the feel impossible to tune coherently.

The measurable test: *grep the components directory for a quoted sentence, a date, a hex colour, a URL, or a bare duration. Zero hits.* That check runs pre-commit, so the rule enforces itself rather than depending on discipline.

03 STACK

Choices, and what I rejected

CONCERN	CHOICE	WHY THIS, NOT THAT
Framework	Next.js 15, App Router, React 19	Your CV claims Next.js — the portfolio should be built in the stack it advertises. Gives RSC, image optimisation, metadata API, and free Vercel deploy. <i>Considered and rejected:</i> Astro would ship less JS, but the credibility of demonstrating your own stated stack outweighs it.
Language	TypeScript, strict	The content schema <i>is</i> the safety net. Without strict mode the whole L1 layer is decorative.
Styling	Tailwind CSS v4	v4's CSS-first @theme makes tokens actual CSS variables — the no-hardcode rule becomes structural instead of aspirational.
Motion	motion (Framer Motion 12)	Declarative orchestration, layout animations, scroll-linked values, and a first-class reduced-motion hook. ~34 KB gz, imported per-section. Rejected GSAP — heavier licence surface for no gain here.
Hero visual	Canvas 2D, hand-written	~6 KB of code replaces a ~500 KB PNG, is crisp at every resolution, themes with your tokens, and actually rotates in 3D. See §06.
Search	Hand-rolled BM25, ~4 KB	Full control over scoring, tokenisation, and the index format. Rejected Fuse.js (fuzzy matching, not relevance ranking) and Lunr (~29 KB for features we don't need).
Validation	Zod	One library validates content, env, and the search index shape. Types are inferred, so schema and TS never drift.
Icons	lucide-react	Tree-shakeable, consistent stroke weight. Icon names come from content, so swapping one is a data edit.
Testing	Vitest + Playwright smoke	The high-value suite is the assistant's question fixture set (§05) — roughly 40 question→expected-answer pairs that run in CI.
Host	Vercel Hobby (free)	Preview deploy per branch, edge CDN, zero config. Nothing is Vercel-locked — with no server route the site can static-export to Cloudflare Pages, Netlify, or GitHub Pages unchanged.

DELIBERATELY NOT INCLUDED

No LLM API — see §05. **No CMS** — you are the only author; a typed file in git is faster and better-reviewed. **No database** — nothing persists. **No auth** — nothing is private. **No smooth-scroll library** (Lenis, Locomotive) — hijacking native scroll breaks trackpad feel, keyboard navigation, and screen readers; the premium feel comes from what happens *during* the scroll, not from replacing it.

04 REPOSITORY Where everything lives

```
faiz-portfolio/
├─ src/
│  ├─ app/
│  │  ├─ layout.tsx · page.tsx      page maps the section registry → components
│  │  ├─ opengraph-image.tsx        OG card rendered from content
│  │  └─ sitemap.ts · robots.ts     no /api directory – nothing server-side
│  ├─ config/
│  │  ├─ env.ts                     the ONLY file that reads process.env
│  │  ├─ motion.ts                  L3 – every duration, spring, stagger
│  │  └─ features.ts · site.ts
│  ├─ content/                      ← L1. every fact about you.
│  │  ├─ schema.ts                  Zod – the contract
│  │  ├─ profile.ts · experience.ts · projects.ts
│  │  ├─ skills.ts · achievements.ts · education.ts
│  │  ├─ vocabulary.ts              synonyms + aliases for the assistant
│  │  ├─ sections.ts                ordered registry → nav, scroll-spy, footer
│  │  └─ index.ts                   validates on import; build fails on bad data
│  ├─ lib/assistant/                ← runs 100% in the browser
│  │  ├─ passages.ts                content → retrievable passages
│  │  ├─ index.build.ts              build step → static BM25 index JSON
│  │  ├─ tokenize.ts                normalise, stem, expand via vocabulary
│  │  ├─ retrieve.ts                 BM25 scoring
│  │  ├─ intents.ts                 ordered intent matchers
│  │  ├─ compose.ts                  matches + content → rich answer blocks
│  │  └─ fixtures.test.ts            ~40 question → expected-topic pairs
│  ├─ lib/visuals/
│  │  ├─ brain.ts                   the rotating neural point-cloud
│  │  └─ wave.ts                    contact-section signal wave
│  ├─ components/
│  │  ├─ sections/                  Hero · Journey · Projects · Assistant · Contact
│  │  ├─ chrome/                   Nav · Footer · ThemeToggle · ScrollProgress · Cursor
│  │  └─ ui/                        primitives – no content, ever
│  └─ styles/globals.css             ← L2. @theme tokens.
└─ scripts/build-index.ts            runs before next build
```

Note what is *absent*: no `app/api/`, no `.env.example` full of secrets, no rate-limit module. Removing the paid API removed an entire tier of the system.

A real answer engine, for exactly zero rupees

You want an assistant that answers questions about your portfolio and costs nothing. Three ways to get there — I evaluated all of them.

APPROACH	REAL COST	VERDICT
Hosted LLM API (Claude, GPT, Gemini paid tiers)	\$8-40 / 1k questions	REJECTED You've ruled out spending. Also needs a server, a key, rate limiting, and abuse monitoring.
Free-tier LLM API (Groq, Google AI Studio, OpenRouter free models)	\$0 today	NOT DEFAULT Free <i>today</i> . Still needs a key, a server route, and it can be throttled or withdrawn without notice. Kept as an optional flag, shipped off.
Browser-local LLM (WebLLM, transformers.js)	\$0	REJECTED 300 MB–1.5 GB model download on first visit. It would destroy your load time and be unusable on mid-range phones — actively damaging the thing it's meant to impress.
In-browser retrieval & answer composition	\$0 — permanently	CHOSEN ~10 KB of code and a ~25 KB index. Answers in under 50 ms, works offline, cannot hallucinate, nothing to rate-limit because there is nothing to abuse.

WHY THIS IS THE BETTER ANSWER, NOT JUST THE CHEAPER ONE

Every portfolio chatbot on the internet is a thin wrapper around someone else's API. An engineer reviewing your site knows that within two questions. A retrieval engine you built — tokeniser, inverted index, BM25 ranking, intent routing, answer composition, with a test suite proving it — is a substantially stronger signal, and it is the one thing on the page a reviewer cannot assume you outsourced.

It also removes the failure mode that would hurt you most: an LLM confidently inventing a job you never had, in front of a recruiter. This engine physically cannot — every sentence it emits is composed from data that passed Zod validation.

HOW IT WORKS

1. **Passages.** At build time, the content layer becomes ~30–40 retrievable passages: one per project, per role, per skill cluster, per achievement, plus profile and contact. Each carries a kind, title, searchable text, entity tags, and a link back to its section.
2. **Index.** A build script tokenises, lightly stems, and produces an inverted index with document lengths for BM25 scoring. Serialised to JSON, ~25 KB gzipped, shipped in the bundle. No network request at runtime.

3. **Query expansion.** The question is normalised and expanded through `vocabulary.ts` — a content-layer alias map, so it's editable data, not code. “tech stack” → technologies; “where do you work” → experience; “ML” → machine learning; “grad” → education.
4. **Intent routing.** An ordered set of matchers classifies the question — project detail, skill check, experience, availability, education, achievements, contact, overview. Each intent owns a composer.
5. **Composition.** The composer builds a *rich* answer from validated data: sentences with real slot-filled values, plus optional chips, a project card, a timeline fragment, or a jump link. Richer than a chatbot's wall of text.
6. **Honest fallback.** If the top BM25 score is below threshold and no intent matches, it says so plainly and offers three things it *can* answer — generated from your content, so the suggestions are never stale.

WORKED EXAMPLES

QUESTION	WHAT HAPPENS
“Tell me about the API monitoring platform”	BM25 top hit → that project passage → project-detail composer → summary, real stack list, two highlights, a jump link to the section.
“Do you know Docker?”	skill-check intent, entity matched against the tech vocabulary → “ <i>Yes — Docker is part of my DevOps toolkit. I’ve used it in the API Monitoring Platform and in production work at Metawurks AI.</i> ”
“Have you worked with Kubernetes?”	Entity not in vocabulary → “ <i>Kubernetes isn’t something I’ve worked with yet. Closest in my stack: Docker and AWS (EC2, Lambda, S3).</i> ” Honest, and still useful.
“What’s your experience?”	experience intent → both roles, real dates, one line each, rendered as a compact timeline fragment.
“Are you open to work?”	availability intent → profile availability line + contact links.
“What’s your favourite film?”	Below threshold, no intent → graceful deflect + three generated suggestions.

MAKING IT FEEL LIKE AN ASSISTANT

The engineering above is invisible; the experience is what a visitor judges. Four cheap details do the work: a brief considered pause (~300 ms) with animated thinking dots; a typewriter reveal of the composed answer at ~18 ms per character, click-to-complete; answer blocks that animate in with a stagger; and suggested questions that shuffle to match whatever section you’re currently scrolled to. All of it is CSS and `requestAnimationFrame` — no cost, no latency.

NAMING IT HONESTLY

The mockup labels this “AI Portfolio Assistant”. Retrieval and ranking are legitimately AI, so the label isn’t false — but do **not** add “powered by GPT/Claude” badges, because that would be. Recommended microcopy: *“Ask anything about my work — answers come straight from my portfolio.”* Accurate, inviting, and the precision itself reads as confidence.

QUALITY GATE

The unusual advantage of a deterministic engine is that it is genuinely testable.

`fixtures.test.ts` holds ~40 question→expected-topic pairs, including deliberate misses and misspellings, and runs in CI. If a content edit breaks retrieval, the build tells you — something no LLM-backed assistant can offer.

06 MOTION & DESIGN

The animated layer, specified properly

You want it to feel high-end and alive. The gap between “advanced” and “janky” is almost entirely discipline, so this is specified as a system rather than a list of effects.

THE HERO BRAIN – THE CENTREPIECE

You asked for the brain to rotate slowly. The honest problem: the image you shared is a flat picture of a 3 D object, and CSS-rotating it looks broken rather than dimensional — the lighting and perspective stay fixed while the shape spins.

Recommendation: generate it instead. A Canvas 2 D neural point-cloud, tuned to resemble the reference:

- ~220 points sampled across a brain-shaped volume (two lobes plus stem), deterministically seeded so it renders identically every load.
- Rotation of ~0.0025 rad per frame around the vertical axis — one full turn in roughly 42 seconds. Slow enough to feel ambient, fast enough to read as alive.
- Perspective projection with depth-driven size and opacity, so the far side genuinely recedes.
- Synapse lines between near neighbours, opacity by depth, capped at ~350 edges.
- A travelling signal pulse firing along a random edge roughly every 800 ms — the detail that makes it read as *neural* rather than decorative.
- The reference’s glowing base platform, as an elliptical ring with a slow rotating gradient sweep.
- $\pm 6^\circ$ damped parallax toward the cursor.

It is roughly 6 KB of code against a ~500 KB image, stays crisp on every display, recolours itself from your design tokens in both themes, and genuinely rotates. The fallback if you’d

rather keep your image: composite it over a rotating ring and give it a slow float and depth parallax — convincing, and about an hour's work instead of half a day. Decision 2 in §08.

FIVE LAYERS OF MOTION

LAYER	WHAT IT DOES
Entrance	One orchestrated page-load sequence: nav settles, headline rises word by word on a 60 ms stagger, brain fades up and starts turning, actions spring in last. Runs once.
Scroll narrative	Each section enters <i>differently</i> — the same fade-up five times is the tell of a generated site. Journey draws its timeline as you scroll; projects wipe in on a clip-path; skill clusters stagger; the contact wave amplifies on approach.
Interaction	Magnetic buttons that lean toward the cursor; link underlines that draw rather than appear; project cards that lift with a light source tracking the pointer; a nav pill that slides between active sections.
Ambient	The brain; a slow gradient-mesh drift behind the hero; the signal wave in contact; a 2–3% grain overlay for depth. All paused by IntersectionObserver when off-screen.
Transition	Native View Transitions for theme switching and any route change — free, progressive, and smoother than a JS crossfade.

THE DISCIPLINE THAT KEEPS IT SMOOTH

- **Compositor only.** Animate `transform` and `opacity`, never `top`, `left`, `width`, or `height`. This single rule is most of the difference between 60 fps and stutter.
- **Nothing animates off-screen.** Every canvas and ambient loop is gated by IntersectionObserver and stops on tab blur.
- **Canvas budget.** DPR capped at 2; particle count scales with viewport (~120 on mobile); glow drawn from a pre-rendered sprite rather than per-point `shadowBlur`, which is the usual cause of canvas frame drops.
- **Zero layout shift.** Every animated element has its space reserved before it animates. Entrances must not move the page.
- **Reduced motion is a real alternative.** One flag from L3 disables the lot — the brain renders a single static frame, reveals become instant, ambient loops never start. Not shortened durations: genuinely still, and genuinely complete.
- **Per-section imports.** Motion code is dynamically imported so the initial route stays under budget.

VISUAL DIRECTION

Keep the reference's instincts — deep ground, luminous accent, oversized type, real whitespace — while avoiding its one AI-tell: the violet-to-blue gradient is the single most over-produced palette in generated design, and a reviewer will have seen it fifty times this month. Proposal: a near-black ground with a faint cool bias, *one* luminous accent carrying every emphasis and

interactive state, an editorial display face against a clean body face, and hairline rules doing the structural work that cards do in the mockup. I'll present the palette and type pairing for your approval at the start of Phase 3, before any section is built.

07 COST LEDGER

Every line item, zeroed

LINE ITEM	MONTHLY	HOW IT'S ZERO
Hosting	\$0	Vercel Hobby. 100 GB bandwidth — a portfolio uses a fraction of a percent. Cloudflare Pages and GitHub Pages are equally free fallbacks.
Assistant	\$0	Runs in the visitor's browser. No API, no key, no server invocation.
Hero visual	\$0	Generated in Canvas. No stock asset, no licence, no CDN.
Fonts	\$0	Self-hosted open-licence faces, subset and served from your own origin. No Google Fonts request.
Database / CMS	\$0	Neither exists. Content is typed files in git.
Analytics	\$0	Vercel Analytics free tier, behind a flag, off by default.
SSL / CDN	\$0	Included with every host above.
Domain (optional)	\$0 or ~\$10/yr	<code>faiz.vercel.app</code> costs nothing. A custom domain is the only spend anywhere in this project, and it is entirely your call. Decision 3 in §08.

Recurring cost with the free subdomain: exactly zero, indefinitely. Nothing in the architecture can generate a bill, because nothing in it is metered — there is no usage-based service anywhere in the stack.

08 BUILD PLAN

Seven phases, each one deployable

PHASE Foundation

0

Repo + git init, Next 15 / TS strict / Tailwind v4, token scaffold, `env.ts`, Vercel project linked.

0.5 d | EXIT CRITERIA

Blank styled page live on a Vercel URL; `npm run build` green.

PHASE **Content layer**

1 Zod schemas for all content types; every résumé fact entered; vocabulary map seeded; section registry; validation wired into build.

0.5 d EXIT CRITERIA

Content validates; a deliberately broken entry fails the build with a useful error.

PHASE **Motion & design system**

2 Palette and type pairing approved, token set complete, `motion.ts` defined, reduced-motion gate, nav from registry, theme toggle with View Transitions, scroll progress, focus states.

1.5 d EXIT CRITERIA

Nav is data-driven; both themes pass WCAG AA; the reduced-motion flag verifiably kills all motion.

PHASE **Hero & the brain**

3 The centrepiece: Canvas neural point-cloud, rotation, synapses, pulses, base ring, cursor parallax, entrance choreography, derived stat counters.

1.5 d EXIT CRITERIA

Steady 60 fps on a mid-range phone; pauses off-screen and on tab blur; renders a clean static frame under reduced motion.

PHASE **Sections 02 · 03 · 05**

4 Journey with a scroll-drawn timeline, Projects with per-card choreography, Contact with the signal wave. Fully content-driven.

2 d EXIT CRITERIA

A complete, animated, shippable portfolio exists. Lighthouse ≥ 95 mobile; zero content literals in `components/`.

PHASE **The assistant**

5 Passages, build-time index, tokeniser, BM25, intents, composers, fixture suite, and the conversational UI with typewriter reveal.

1.5 d EXIT CRITERIA

All ~40 fixtures pass; unknown questions deflect gracefully; works with the network disconnected.

PHASE Ship

6

OG image, sitemap, robots, JSON-LD person schema, analytics flag, domain, README runbook, cross-browser and device pass.

0.5 d

EXIT CRITERIA

Link preview renders on LinkedIn and WhatsApp; a stranger can clone and run it with no configuration at all.

Total ≈ 8 working days — one more than Rev. A: the assistant got simpler, but the motion work grew by more than that. Phases 0–4 (≈6 days) get you a portfolio you could send to a recruiter today.

09 OPEN DECISIONS

Four things left for you

Down from six — confirming the mockup was placeholder resolved the email, the stats, and the phantom project outright.

1. Skill bars, or capability clusters?

Recommendation: clusters with evidence links. Self-assigned percentages read as decoration to an engineer; “RAG, agentic AI, MCP → see this project” is verifiable. The schema supports both, so it’s reversible — but it changes the composition of section 02, so decide before Phase 4.

2. Generated brain, or your image plus a rotating ring?

Recommendation: generated. It’s the difference between something that genuinely rotates and something that visibly spins a flat picture — and it’s 6 KB against 500 KB. Costs about half a day more. Your image stays as the fallback if you prefer it.

3. Free subdomain, or a custom domain?

`faiz.vercel.app` is free forever and perfectly respectable. A domain like `mdfaiz.dev` is roughly ₹800–1,200 a year and does read more established on a résumé. **The only possible cost in the whole project** — entirely optional, and easy to add later without changing anything.

4. Blog: build the shell, or leave it out?

Recommendation: build it behind a flag, ship with the flag off. About two hours now, saves a refactor when you have your first post. An empty blog with a “coming soon” is worse than no blog at all.

ALSO USEFUL, NOT BLOCKING

Live URLs and repository links for the four projects, if they exist — a “view source” link is worth more than any description. And your LinkedIn and GitHub URLs, which the CV references but doesn’t spell out.

10 GATES

What “done” is measured against

GATE	TARGET	WHY IT’S THE TARGET
LCP (mobile, 4G)	< 2.0 s	An AI engineer’s portfolio that loads slowly undercuts its own argument.
CLS	< 0.05	Entrance animations are the usual culprit — reserve space before animating.
Hero frame rate	60 fps	Measured on a mid-range Android, not a desktop. A stuttering brain is worse than no brain.
Lighthouse a11y	100	Non-negotiable. Keyboard-reachable, visible focus, semantic landmarks, real alt text.
Assistant response	< 50 ms	Local computation. Any perceived delay is the deliberate 300 ms pause, not the engine.
Assistant fixtures	40 / 40	The quality gate a deterministic engine makes possible.
Content literals in components/	0	The no-hardcode rule, made checkable. Enforced pre-commit.
Total JS (initial route)	< 200 KB gz	Motion is the main risk; import it per-section, not globally.
Works fully offline after first load	yes	Proves nothing in the experience depends on a paid service.

PRINCIPAL RISKS

- **Animation jank on mid-range phones — now the highest-impact risk.** An ambitious hero that stutters is worse than a modest one that doesn’t. Mitigated by the Phase 3 exit criterion being measured on real hardware, and by the compositor-only rule.

- **Assistant brittleness.** A deterministic engine handles unusual phrasings less gracefully than an LLM. Mitigated by the vocabulary alias map being content-layer data you can extend in seconds, and by a fallback that always offers something useful rather than failing.
- **Over-animation.** Too much motion is the fastest route to a site reading as AI-generated. Mitigated by the five-layer budget in §06 being written down before we start, and by varying the choreography per section rather than repeating one reveal.
- **Generic palette.** The reference's violet gradient is the most common generated look there is. Mitigated by an explicit direction review in Phase 2, before any section is built.